Republic of the Philippines

Bicol University - Polangui

Computer Studies Department

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IS ELEC 2 | WEB SYSTEMS AND TECHNOLOGIES

PHASE I - FOUNDATION OF FINAL PROJECT

BU EntrepHub: Integrated Food Stall Directory and Digital Menu System.

GROUP 2

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PROJECT TITLE: BU EntrepHub: Integrated Food Stall Directory and Digital Menu System

INTRODUCTION & SYSTEM PURPOSE

During short lecture breaks, Bicol University Polangui students encounter long queues, tight schedules, and uncertainty regarding open food stalls or available menu items. BU EntrepHub is a web-based platform bridging student consumers and student entrepreneurs. It provides a real-time directory to check stall availability, compare prices, and filter budget-friendly meals before walking across campus, while giving student vendors a simple dashboard to manage menus and daily operating statuses.

TARGET USERS

Primary Users:

- **BUP Students:** Browse menus, view real-time availability, filter meals under ₱60, and save time during breaks.
- **Student Entrepreneurs:** Manage stalls for free, update menu items and prices instantly, and toggle status (**Open/Closed**, **Available/Sold Out**).

Secondary Users: BUP Faculty & Staff, Campus Guests/Visitors, and BUP Administration overseeing student entrepreneurship initiatives.

KEY FEATURES

1. Campus Food Feed: Real-time status pills (Open/Closed), budget/category filters (*Rice Meals under ₱60*, *Snacks & Drinks*), GCash badges, and price ranges.

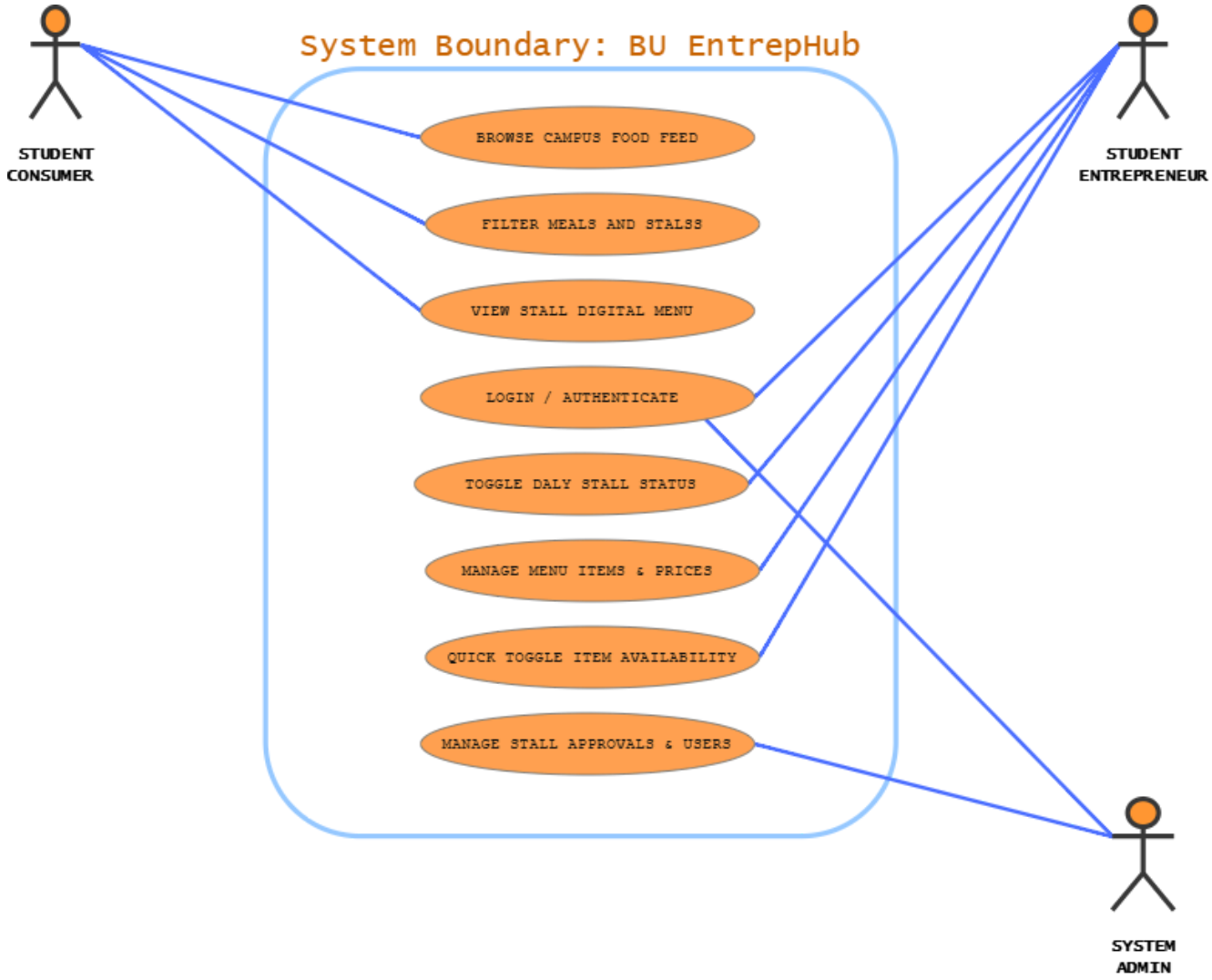
2. Stall Digital Menu Page: Displays stall section, operating hours, and categorized menu items with Bestseller and Available/Sold Out badges.

3. Seller Management Dashboard: Authenticated control panel to toggle daily stall status, update pricing, and perform quick item availability switches.

4. Role-Based Access: Secure authentication for student consumers, sellers, and system administrators.

USE CASE DIAGRAM (UML)

Tool used: draw.io



ENTITY-RELATIONSHIP DIAGRAM (ERD)

COLLECTION	ATTRIBUTE	DATA TYPE	CONSTRAINT	DESCRIPTION
USERS	`_id`	*ObjectId*	Primary Key (PK)	Unique user identifier
	`name`	*String*	Required	Full name of user
	`email`	*String*	Unique, Required	BUP student/staff email
	`password`	*String*	Required	Hashed account password
	`role`	*String*	Enum	“Consumer”, “seller”, or “admin”
	`createdAt`	*Date*	Default: Now	Registration timestamp

Stalls	`_id`	*ObjectId*	Primary Key (PK)	Unique stall identifier
	`ownerId`	*ObjectId*	Foreign Key (FK)	References USers._id
	`stallName`	*String*	Required	Official stall title
	`section`	*String*	Required	Student section
	`operatingHours`	*String*	Required	Display hours
	`isOpen`	*Boolean*	Default: true	Real-time stall status
	`gcashAvailable`	*Boolean*	Default: false	Accepts GCash flag

MenuItems	`_id`	*ObjectId*	Primary Key (PK)	Unique food item identifier
	`stallId`	*ObjectId*	Foreign Key (FK)	References Stalls._id
	`itemName`	*String*	Required	Name of food item
	`category`	*String*	Required	Foods that serves
	`price`	*Number*	Required	Price in PHP
	`isAvailable`	*Boolean*	Default: true	Stock availability statu
	`isBestseller`	*Boolean*	Default: False	Highlight badge flag

I. RELATIONSHIPS EXPLANATIONS

- **Users → Stalls (**1: 1 **Relationship):** Each user registered under the "seller" role is linked to exactly one food stall via the ownerId field.
- **Stalls → MenuItems (**1: *N* **Relationship):** Each food stall can list multiple menu items linked back to the stall via the stallId field.

SYSTEM ARCHITECTURE DIAGRAM

BU EntrepHub operates on a 3-Tier Model-View-Controller (MVC) web system architecture where frontend user views interface with backend Express routing services and persistent database collections:

1. **Frontend HTML/Bootstrap Views:**

- *index.html* (Food Directory): Renders the campus food stall feed and search bar. Sends *GET /api/stalls* requests to retrieve open stalls and active menus.
- *stall-details.html* (Digital Menu View): Renders specific stall information and item lists. Sends *GET /api/stalls/:id/menu* requests.
- *login.html* (Authentication Page): Renders credential input forms for Student Entrepreneurs and Admins. Sends *POST /api/auth/login requests.*
- *dashboard.html* (Seller Control Panel): Displays toggle switches for stall opening status and item stock availability. Sends *PUT /api/stalls/statu*s an*d POST /api/menu/add requests.*

2. **Application Tier (Node.js & Express.js):**

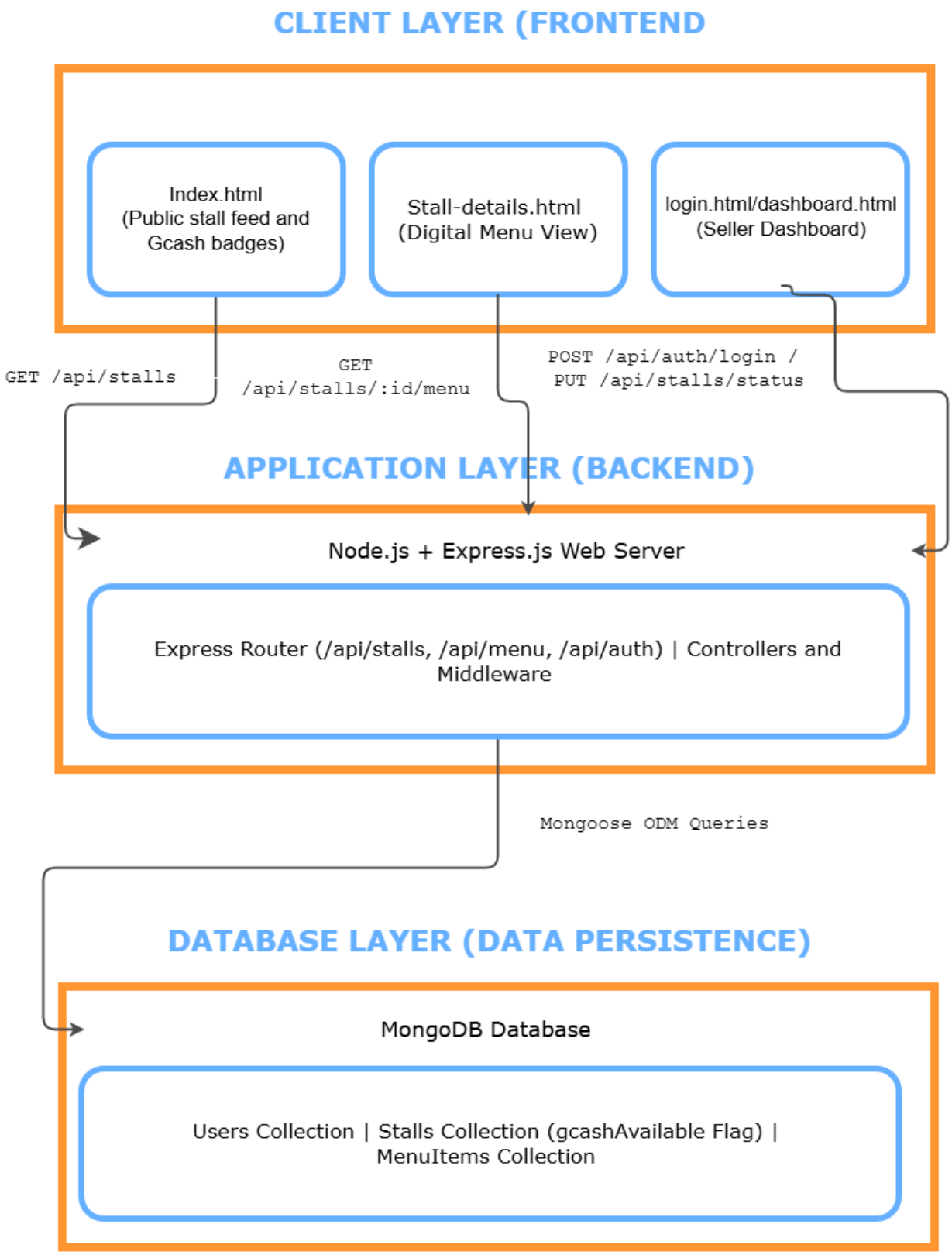
- Acts as the web server handling routing via /api/stalls, /api/menu, and /api/auth.
- Processes request validation, session security, and business logic.

3. **Database Tier (MongoDB & Mongoose):**

- Stores persistent collections for Users, Stalls, and MenuItems.
- Executes database CRUD operations sent from Express controller functions.

VISUAL REPRESENTATION

Tool: draw.io



GITHUB REPOSITORY URL:

https://github.com/brianvielvasquezborje25-netizen/final-project-Group2-BSIS2B

